CSIS 402 : COMPUTER ORGANIZATION & SYSTEM PROGRAMMING

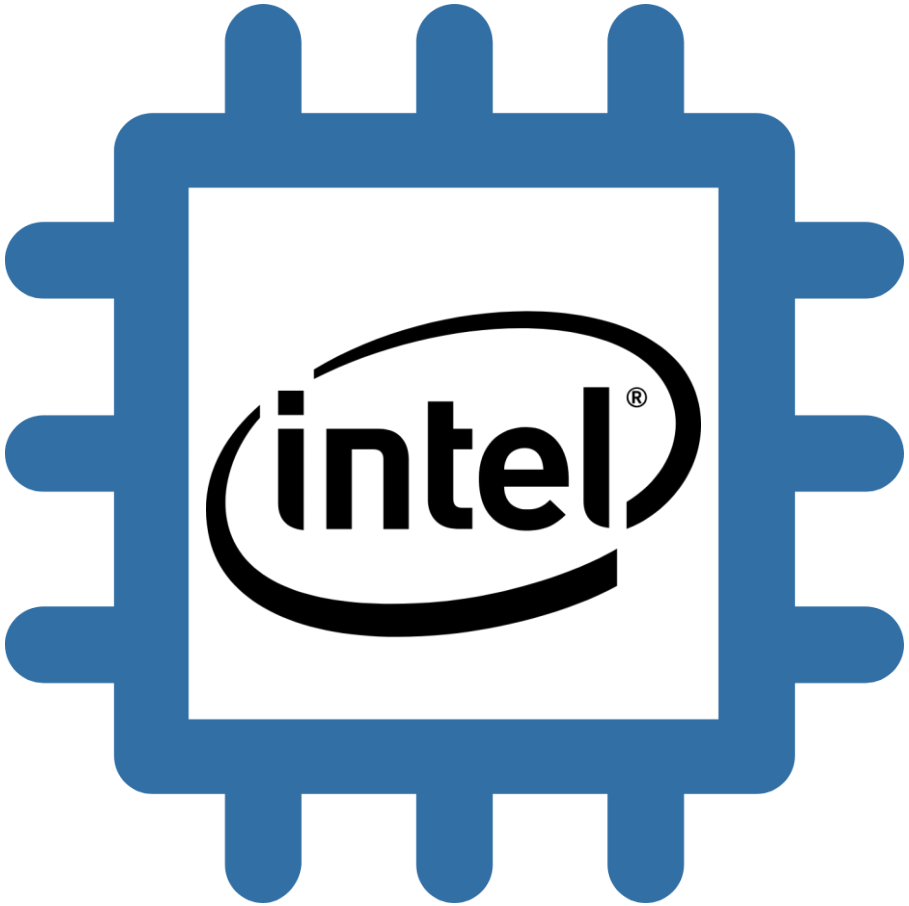
INTEL 80X86 INTERNAL STRUCTURE

LECTURE 5

Dr. Sarah Azzam

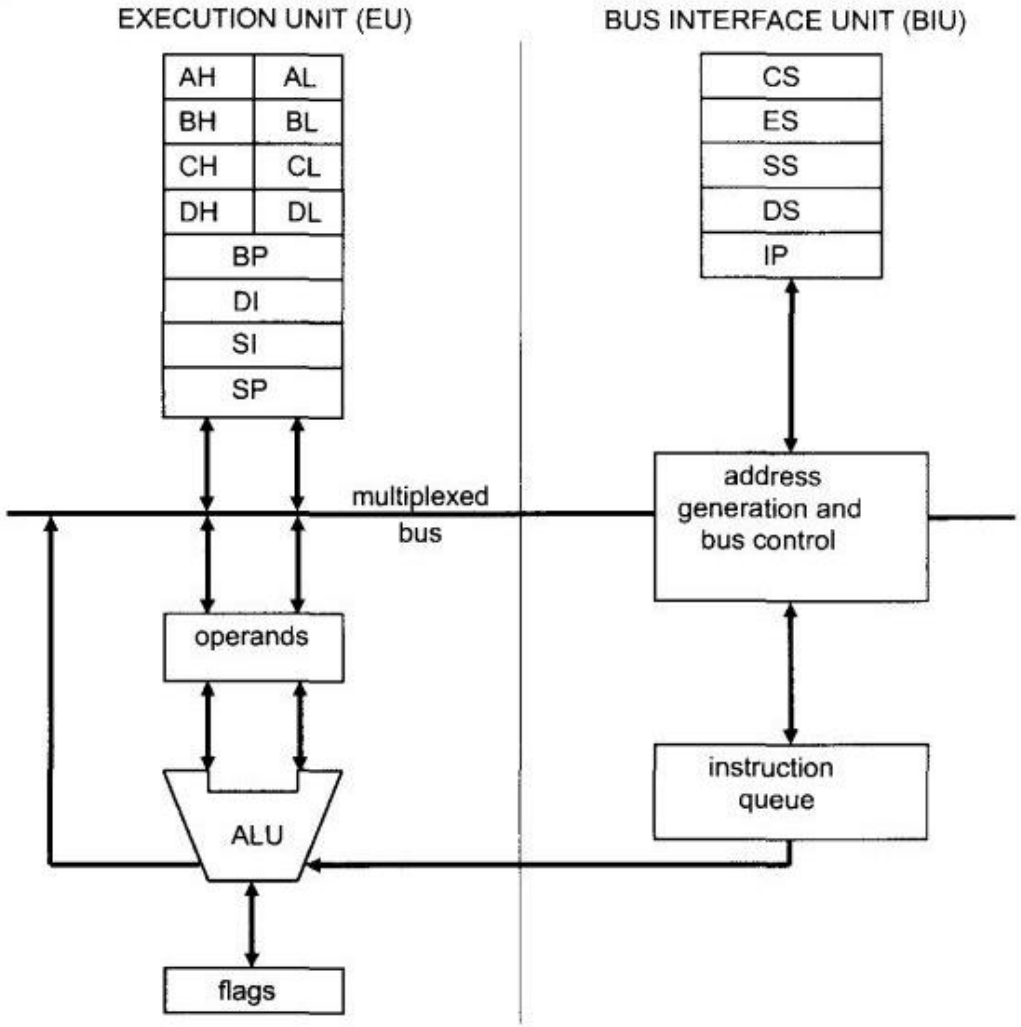
Slides Based on Lecture Slides by Assoc. Prof. Dr. Amr Talaat

GUC
German University in Cairo



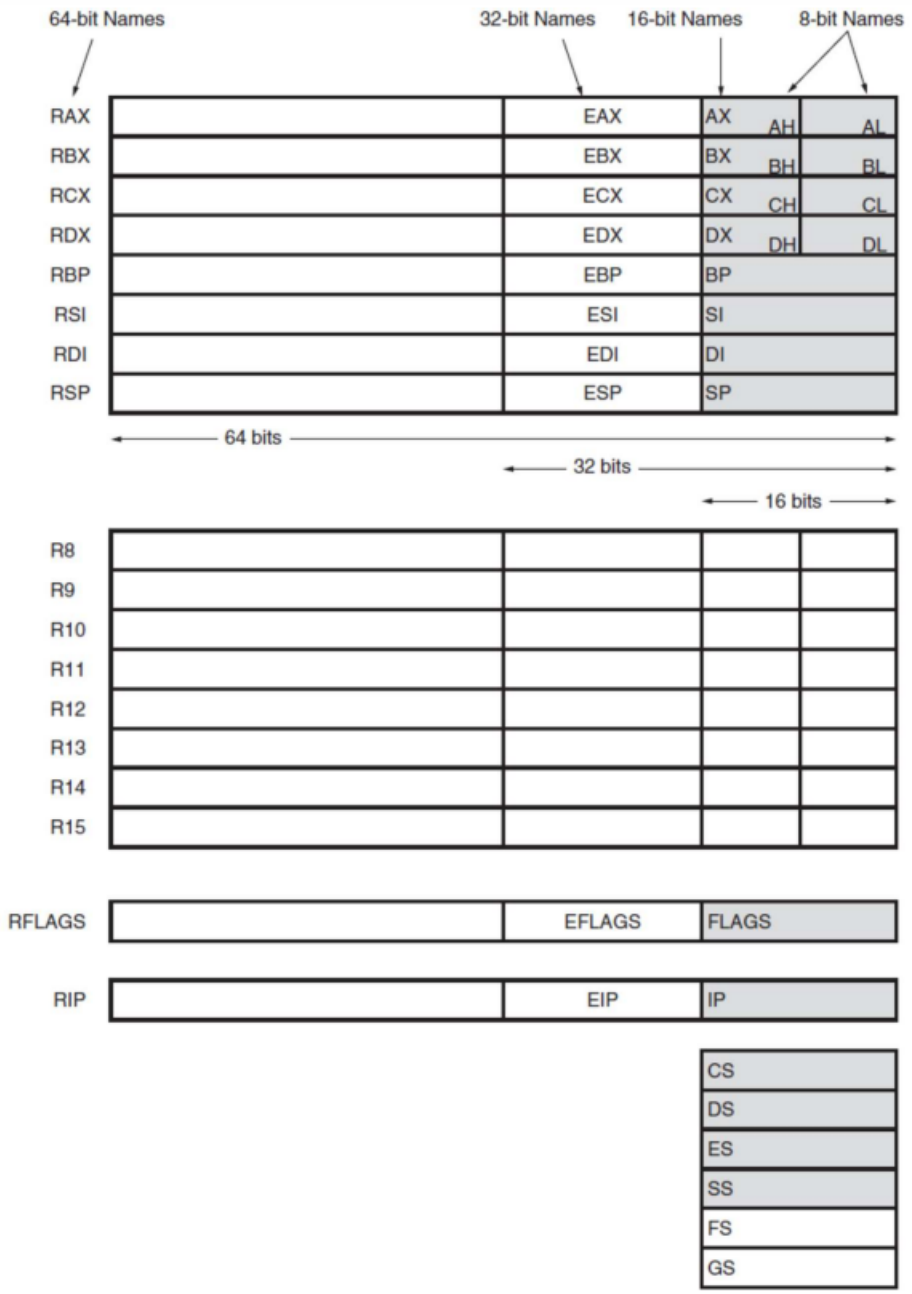
INTEL 80x86 MICROPROCESSOR

INTERNAL STRUCTURE OF THE 8086 MICROPROCESSOR

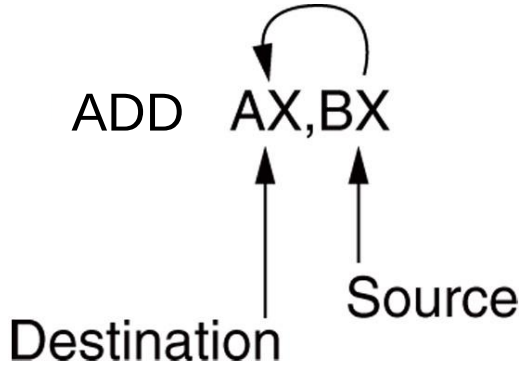


REGISTERS

- Different names are given for 64-bit, 32-bit, 16-bit and 8-bit version of the same register.



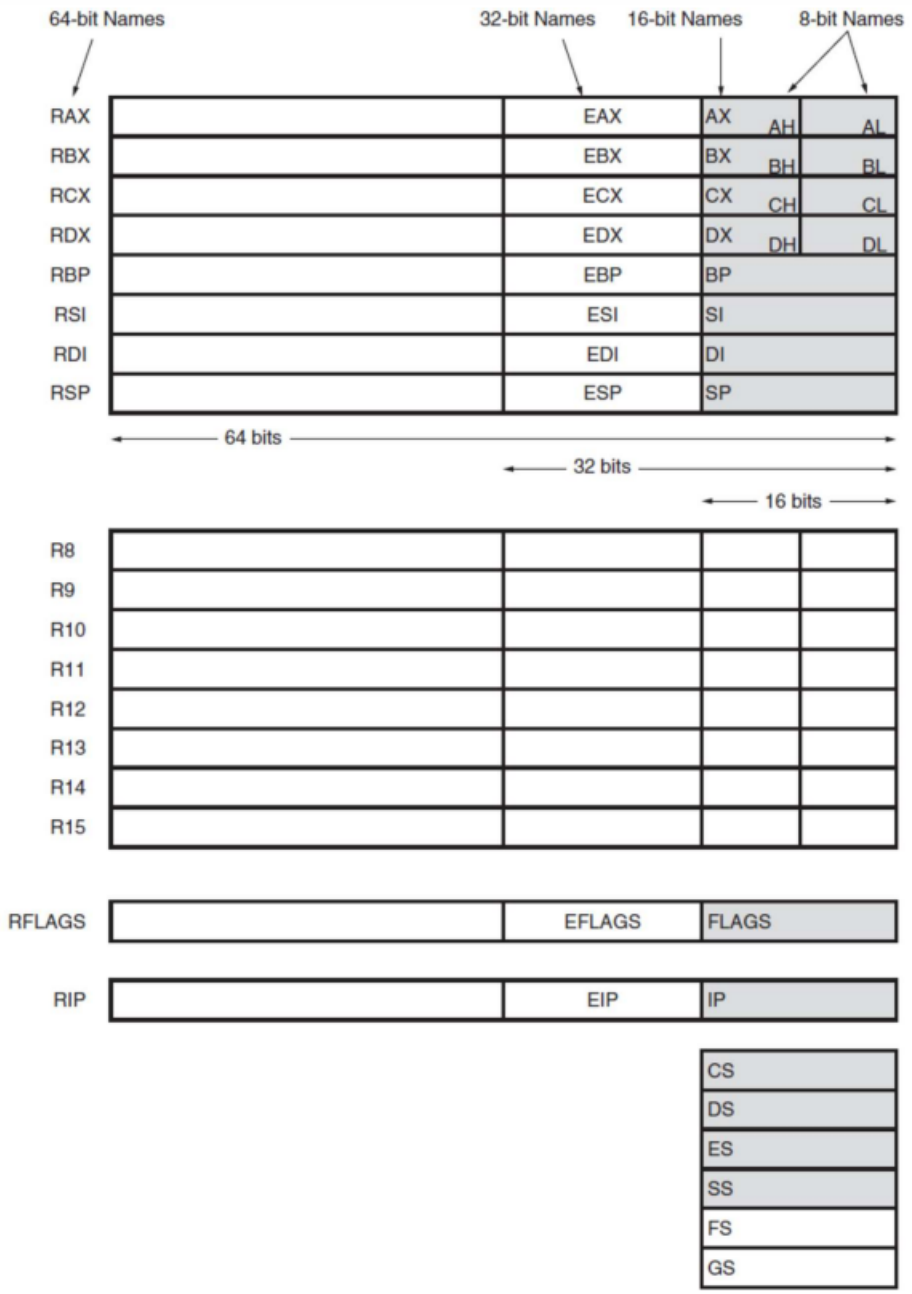
EXAMPLES



1
- Add AL, AH
- The instruction adds the 8-bit contents of AH to AL.
- Only the contents of AL changes based on the previous instruction

2
- Add DX, CX
- The instruction adds the 16-bit contents of CX to DX.
- Only the contents of DX changes based on the previous instruction

3
- Add ECX, EBX
- The instruction adds the 32-bit contents of EBX to ECX.
- Only the contents of ECX changes based on the previous instruction



GENERAL PURPOSE REGISTERS

RAX, EAX, **AX (AH & AL)**

Accumulator

- A general purpose register
- Also an accumulator – stores intermediate results after arithmetic and logic operations

RBX, EBX, **BX (BH & BL)**

Base Index

- A general purpose register
- The BX register sometimes holds the offset address of a location in the memory in all microprocessor versions

GENERAL PURPOSE REGISTERS

RCX, ECX, **CX (CH & CL)**

Count

- A general purpose register
- Also a count register – holds the count for various instructions (e.g., loop count in iterative operations)

RDX, EDX, **DX (DH & DL)**

Data

- A general purpose register
- Also a data register – stores data related to accumulator's calculation (multiply and divide)

GENERAL PURPOSE REGISTERS

RBP, EBP, **BP**

Base pointer

- A general purpose register
- Also a base pointer register – points to a memory location for memory data transfer

RDI, EDI, **DI**

Destination Index

- A general purpose register
- Also a destination index register – holds the memory address for the destination data of a string instruction

GENERAL PURPOSE REGISTERS

RSI, ESI, SI

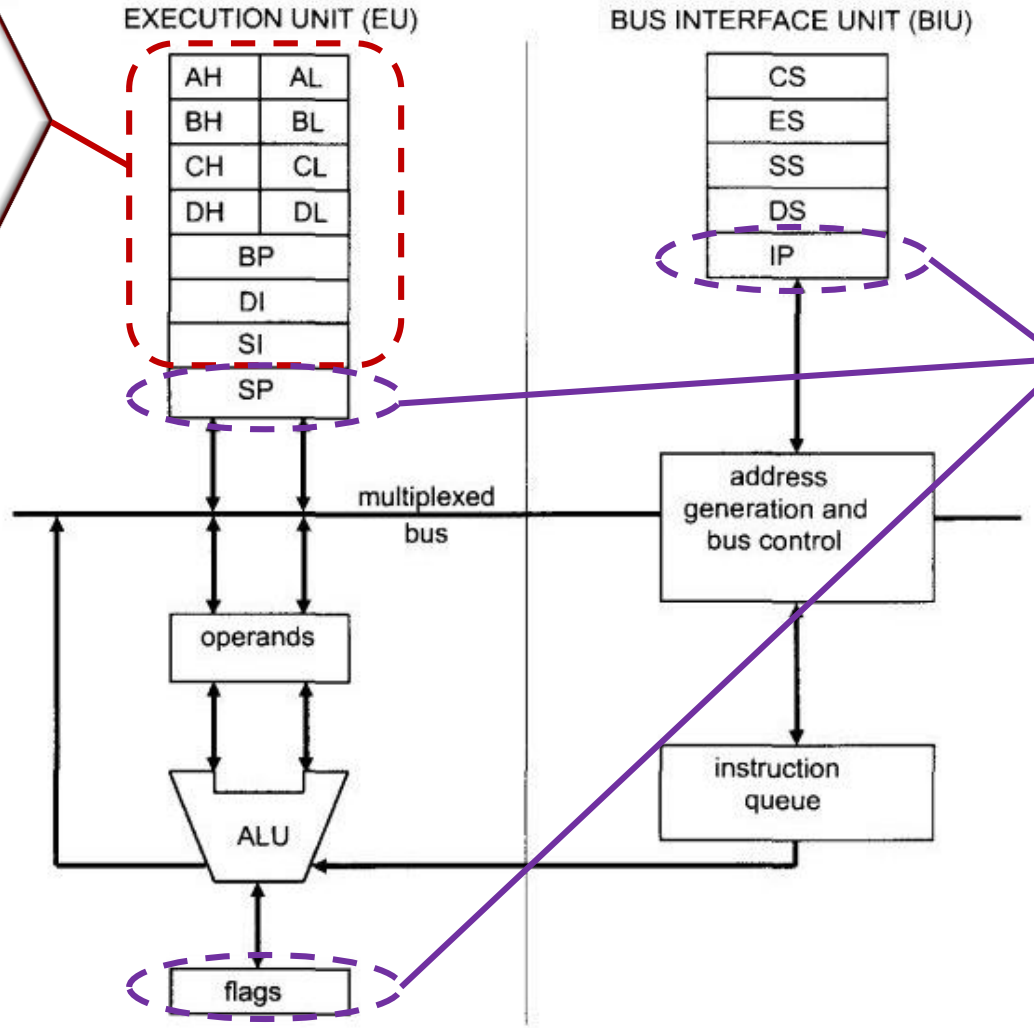
Source Index

- A general purpose register
- Also a source index register - holds the memory address for the source data of a string instruction

R8 through R15

- General purpose registers
- Found only in 64-bit microprocessors
- Data are addressed in 64-, 32-, 16-, or 8-bit sizes

General Purpose/ Multipurpose Registers



Special-Purpose Registers

SPECIAL PURPOSE REGISTERS

RIP, EIP, IP

- Instruction pointer – points to the next instruction in the memory to be executed (i.e., stores the offset address of the next instruction to be executed)

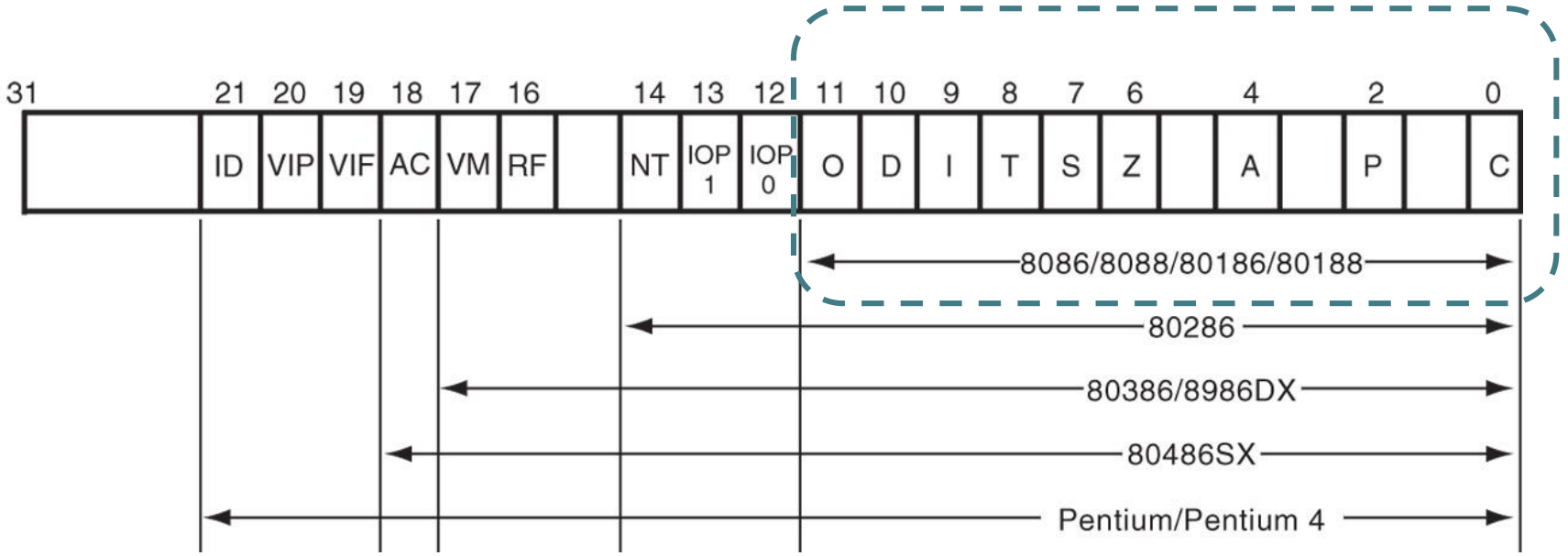
RSP, ESP, SP

- Stack pointer – points to an area in memory called the stack (i.e., provides the offset value within the program stack.)

RFLAGS, EFLAGS, FLAGS

- Indicates the condition of the microprocessor and control its operation

FLAGS


31
21
20
19
18
17
16
14
13
12
11
10
9
8
7
6
4
2
0
ID
VIP
VIF
AC
VM
RF
NT
IOP 1
IOP 0
O
D
I
T
S
Z
A
P
C
8086/8088/80186/80188
80286
80386/8986DX
80486SX
Pentium/Pentium 4

FLAGS

Flag:					O	D	I	T	S	Z		A		P		C
Bit no:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

- The C, P, A, Z, S and O flags are changed by most **arithmetic and logic operations**.
- Flags never change for any data transfer or program control operations.
- Some flags are also used to control features found in the microprocessor.

FLAGS

Flag:					O	D	I	T	S	Z		A		P		C
Bit no:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Descriptions for some of the flag bits:

C (carry): holds the carry after addition or borrow after subtraction (either from bit position 7 in an 8-bit operation or from bit position 15 after a 16-bit operation).

P (parity): the parity is set to one if the low order byte has even number of ones (zero ones is considered even number of ones)

A (auxiliary carry): holds the half-carry after addition or the borrow after subtraction between bit positions 3 and 4 of the result.

FLAGS

Flag:					O	D	I	T	S	Z		A		P		C
Bit no:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

- **Z (zero):** shows that the result of an arithmetic or logic operation is zero.
  - Z = 1 means that the result is zero however, Z = 0 means that the result is not zero.
- **S (sign):** holds the arithmetic sign of the result after an arithmetic or logic instruction executes.
  - S = 0 for positive, S = 1 for negative.
- **O (overflow):** indicates that the result of an addition/subtraction of signed numbers exceeded the capacity of the machine.

FLAGS

Flag:					O	D	I	T	S	Z		A		P		C
Bit no:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0



To be discussed with Interrupts

EXAMPLE

Perform the following binary addition in 2's complement arithmetic. Determine whether there is any overflow. If there is an overflow, then discuss how to avoid it.

(i) $-7 + 10$

(ii) $-10 + 7$

(iii) $10 + 7$

(iv) $-10 + -7$

EXAMPLE

(i) $-7 + 10$



No overflow ($C_5 \oplus C_4 = 0$)

Since $3 \in [-2^{n-1}, 2^{n-1} - 1]$

$$3 \in [-2^4, 2^4 - 1]$$

n → number of bits

OF =0

		$C_5 = 1$	$C_4 = 1$	$C_3 = 0$	$C_2 = 0$	$C_1 = 0$	$C_0 = 0$
-7	=	1	1	0	0	1	
10	=	0	1	0	1	0	
		1	0	0	0	1	1

Discard 

EXAMPLE

(ii) $-10 + 7$



No overflow ($C_5 \oplus C_4 = 0$)

Since -3 $\in [-2^{n-1}, 2^{n-1} - 1]$

-3 $\in [-2^4, 2^4 - 1]$

n → number of bits

OF =0

		$C_5 = 0$	$C_4 = 0$	$C_3 = 1$	$C_2 = 1$	$C_1 = 0$	$C_0 = 0$
-10	=	1	0	1	1	0	
7	=	0	0	1	1	1	
		0	1	1	1	0	1

Discard 

EXAMPLE

(iii) 10 + 7



Overflow ($C_5 \oplus C_4 = 1$)

Since $17 \notin [-2^{n-1}, 2^{n-1} - 1]$

$$17 \notin [-2^4, 2^4 - 1]$$

n → number of bits

OF =1

		$C_5 = 0$	$C_4 = 1$	$C_3 = 1$	$C_2 = 1$	$C_1 = 0$	$C_0 = 0$
10	=	0	1	0	1	0	
7	=	0	0	1	1	1	
		1	1	0	0	0	1

Discard 

EXAMPLE

(iv) $-10 \; - 7$



Overflow ($C_5 \oplus C_4 = 1$)

Since -17 $\notin [-2^{n-1}, 2^{n-1} - 1]$

-17 $\notin [-2^4, 2^4 - 1]$

n → number of bits

OF =1

		$C_5 = 1$	$C_4 = 0$	$C_3 = 0$	$C_2 = 0$	$C_1 = 0$	$C_0 = 0$
-10	=	1	1	0	0	1	
- 7	=	1	0	1	1	0	
		1	0	1	1	1	1

Discard 

EXAMPLE 2

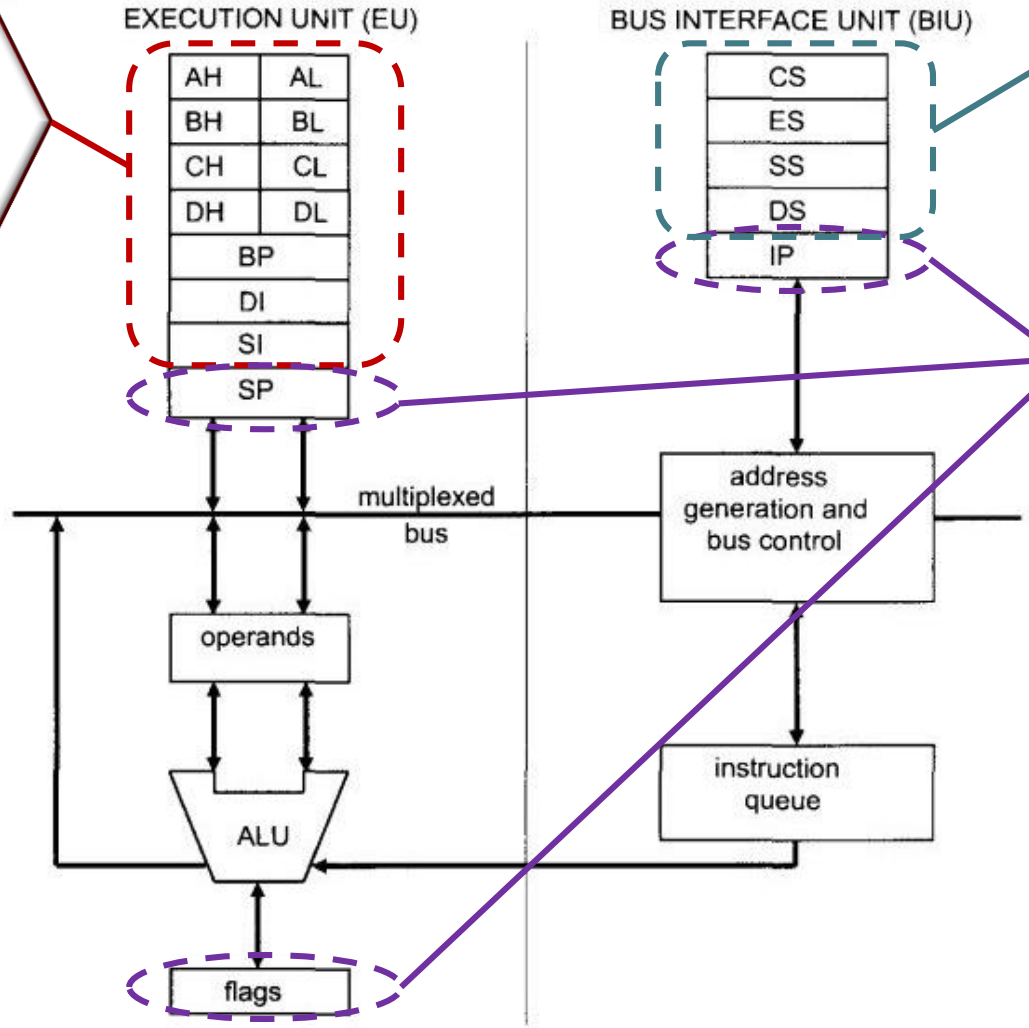
•Show the **flag status** after the following operation
01001111+01110010

P (parity): the parity is set to one if the low order byte has even number of ones (zero ones is considered even number of ones)

A (auxiliary carry): holds the half-carry after addition or the borrow after subtraction between bit positions 3 and 4 of the result.

OF	SF	ZF	AF	PF	CF
1	1	0	1	0	0

General Purpose/ Multipurpose Registers



Segment Registers

Special-Purpose Registers

PROGRAM SEGMENTS

Code Segment

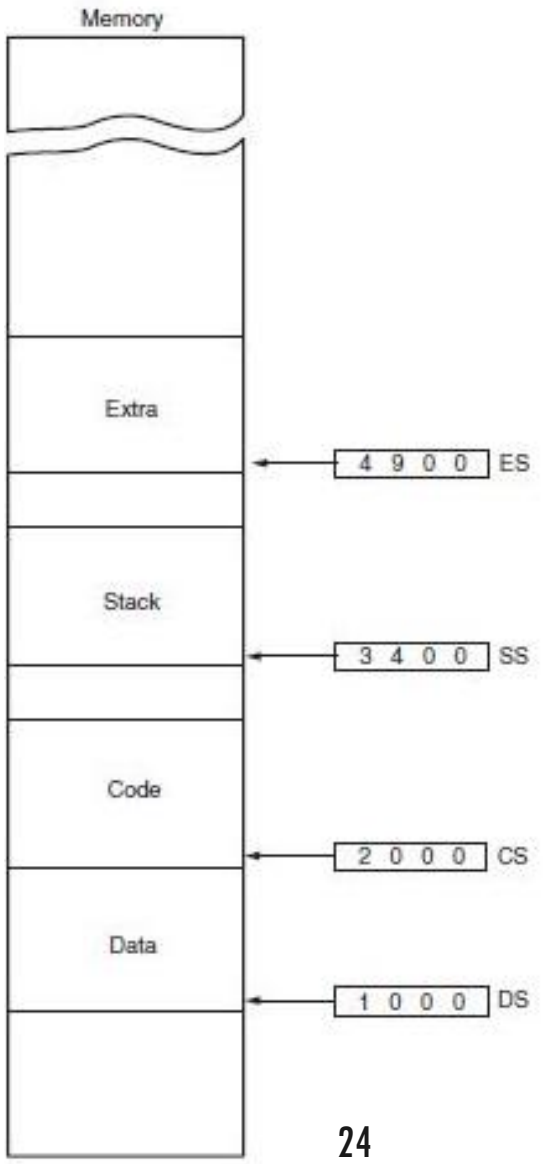
Code

- A section in the memory that holds the code (programs and procedures) used by the microprocessor.
- **Code Segment (CS) register** stores the starting address of the memory section that holds the code.

Data Segment

Data

- A section in the memory that contains most data used by a program.
- **Data Segment (DS) register** stores the starting address of data segment.



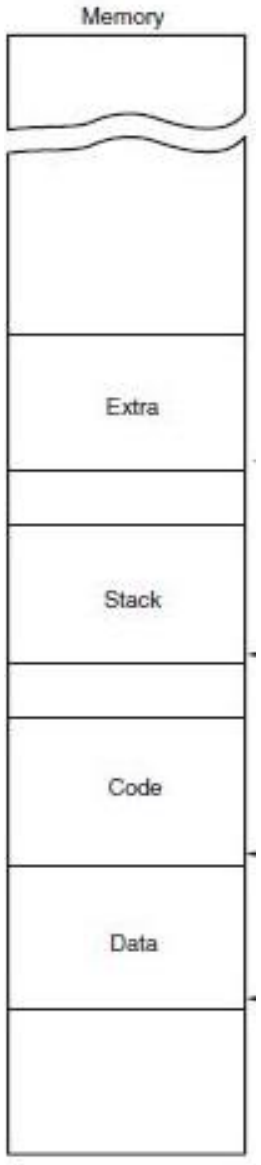
PROGRAM SEGMENTS

Stack Segment

- A section in the memory that stores information temporarily.
- **Stack Segment (SS) register** stores the starting address of the stack segment.

Extra Segment

- A section in the memory that acts as an extra data segment.
- **Extra Segment (SS) register** stores the starting address of the extra segment.

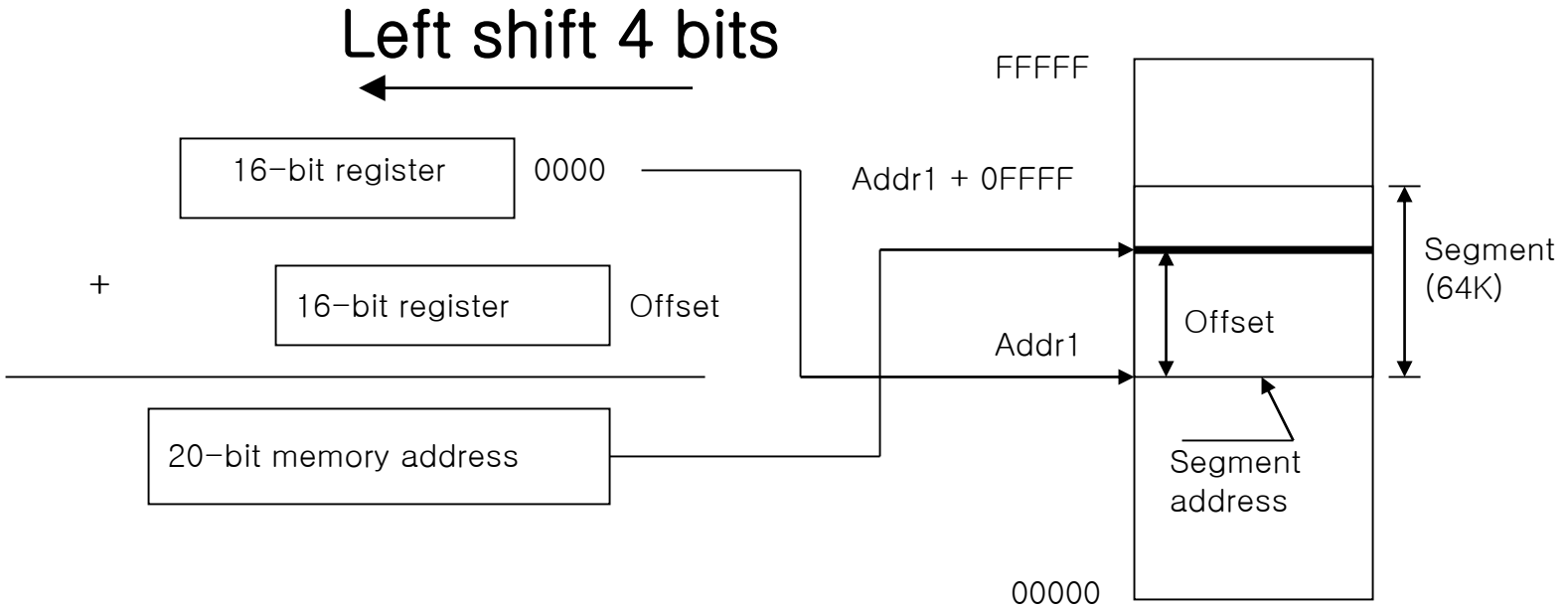


INTEL 8086 MEMORY

- Intel 8086 memory has 1MB memory.
- Intel 8086 memory is byte addressable; each byte has a unique address.

GENERATING MEMORY ADDRESSES

- How can a 16-bit microprocessor generate 20-bit memory addresses?

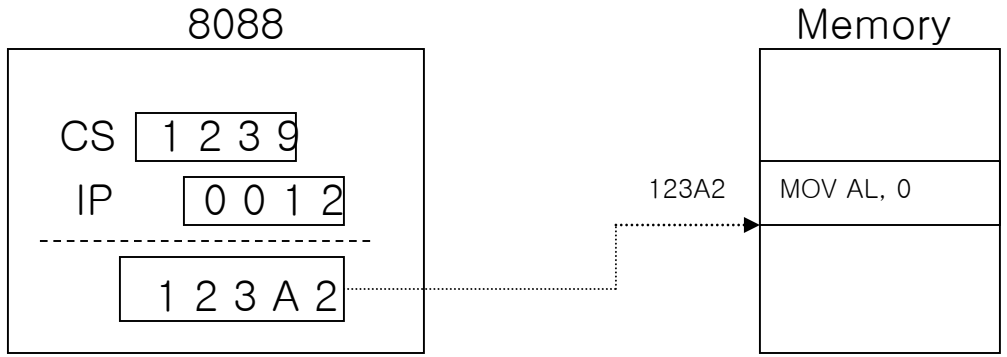


Intel 80x86 memory address generation

1M memory space

FETCHING INSTRUCTIONS

- Where to fetch the next instruction?



Compute the
a) logical address
b) Offset address
c) Physical Address
d) Lower range of the code segment
e) Upper range of the code segment

Answer

a) The logical address → 1239:0012
b) The offset address → 0012
c) The physical address → 123A2
d) The lower range → 12390+0000= 12390
e) The upper range of the code segment → 12390+FFFF=2238F

EXAMPLE

If CS = 3499H and IP = 2500H.
i) Find the logical address.
ii) Determine the offset address.
iii) Calculate the physical address.
iv) Calculate the lower range of the code segment.
v) Calculate the upper range of the code segment.

The logical address → 3499:2500
The offset address → 2500
The physical address → 36E90
The lower range → 34990
The upper range of the code segment → 4498F

REFERENCE TEXTBOOK

[1] “80X86 IBM PC and Compatible Computers: Assembly Language, Design, and Interfacing” by Muhammad Ali Mazidi , Janice Mazidi Gillispie and Janice Gillispie Mazidi

[2] “The Intel microprocessors 8086/8088, 80186/80188, 80286, 80386, 80486, Pentium, Pentium Pro processor, Pentium II, Pentium III, Pentium 4, and Core2 with 64-bit extensions: architecture, programming, and interfacing” by Barry B. Brey”

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[1] https://www.ee.hacettepe.edu.tr/~alkar/ELE336/w2-hacettepe[2016].pdf
[2] https://microcontrollerslab.com/8086-microprocessor-architecture/